

**MCSL-222 (Set-1)**  
**MASTER OF COMPUTER APPLICATIONS**  
**(MCA-NEW)**  
**OOAD and Web Technologies Lab**

Duration : 2 hours

Maximum Marks : 50

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- Note:
1. There are two sections in this paper.
  2. Each section is of one hour duration.
  3. Attempt only that section(s) in which you are not yet successful.

**SECTION-1: OOAD**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

1. Draw collaboration diagram for online Banking System.

**SECTION-2: Web Technologies**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

2. Create a user table into the database and bind the user entity with Spring Security for Login.

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**MCSL-222 (Set-2)**  
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Duration : 2 hours

Maximum Marks : 50

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- Note:
1. There are two sections in this paper.
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**SECTION-1: OOAD**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

1. Draw sequence diagram for Online Examination System. 20

**SECTION-2: Web Technologies**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

2. Create a form to capture customer information (for an online shopping portal) using spring form tags. Write program to bind form objects with entity beans in spring MVC.

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**MCSL-222 (Set-3)**  
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- Note:
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**SECTION-1: OOAD**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

1. An online food delivery company provides a range of facilities to customers through its Mobile App/Web portal. It provide options to choose Restaurant, Food items, Online and COD (Cash on delivery) options, Order Status, and Feedback facility. Perform the following for this system:
  - (i) Identify classes and their associations.
  - (ii) Draw Class Diagram for the System.

## **SECTION-2: Web Technologies**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

2. Write a program using JSP and JDBC create CRUD (Create/Save, Read, Edit/Update, Delete) application for students attendance management in MCSL-222 counselling classes.

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**MCSL-222 (Set-4)**  
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Duration : 2 hours

Maximum Marks : 50

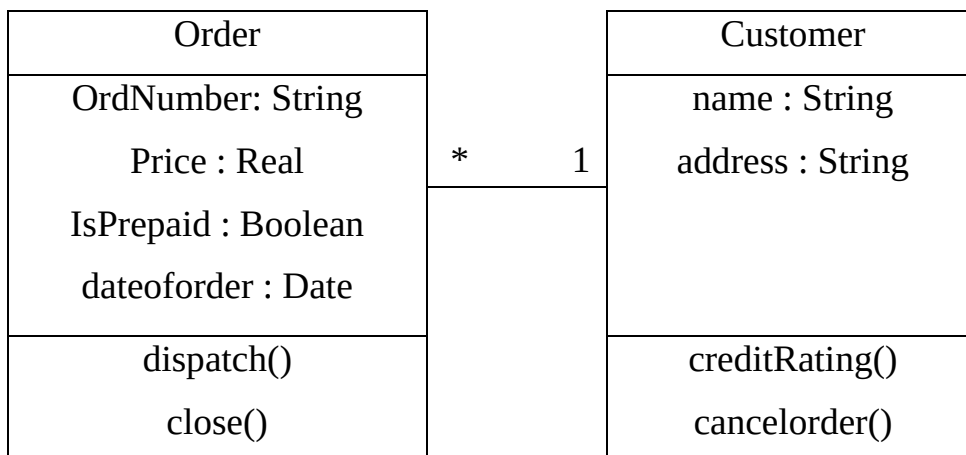
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- Note:
1. There are two sections in this paper.
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**SECTION-1: OOAD**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

1. Implement following association in C++/Java.



## **SECTION-2: Web Technologies**

Note: Answer the following question. It carries 20 marks and the viva-voce is of 5 marks.

2. Write simple program for role-based authentication using Spring Boot.

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